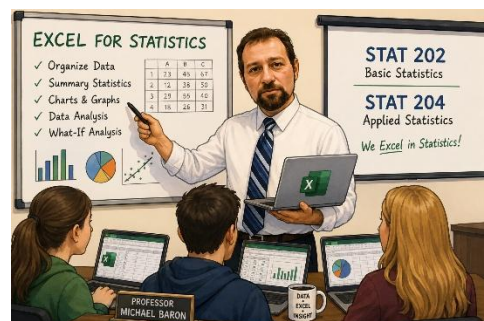
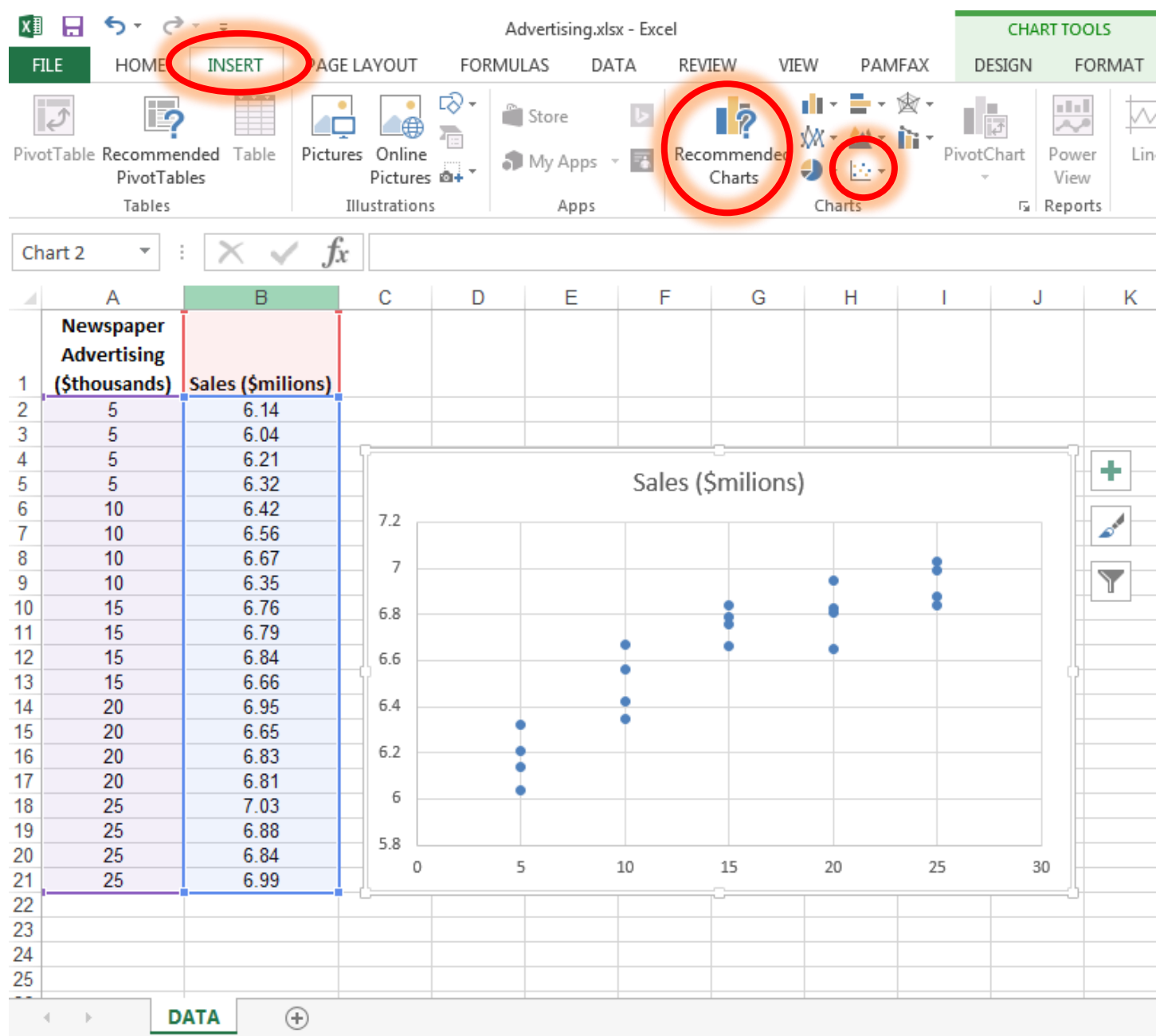


Regression and Correlation

Scatterplots



Insert -> Charts -> Choose a scatterplot that you like.



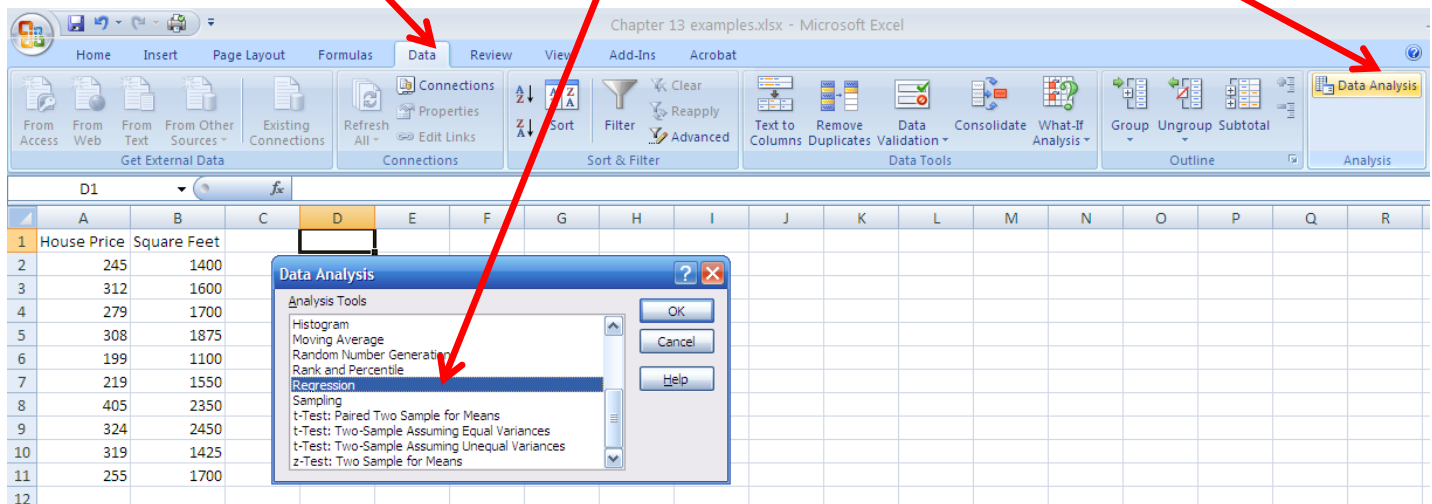
Regression

The fastest way is with the Analysis Toolpak:

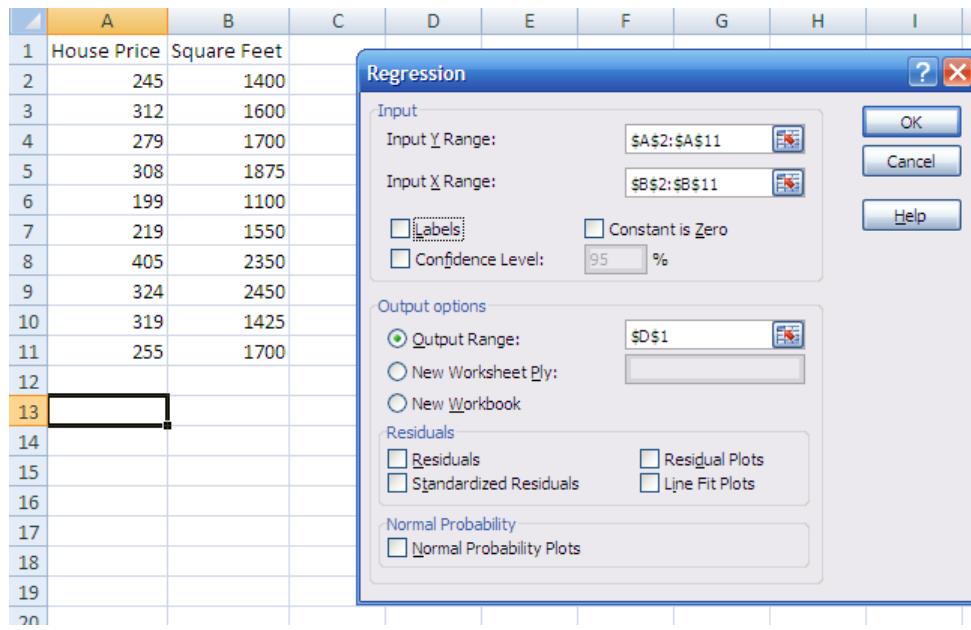
1. Choose Data

2. Choose Data Analysis

3. Choose Regression



Enter the Y range and the X range and the desired options.



Read the output.

<i>Regression Statistics</i>	
Multiple R	0.76211
R Square	0.58082
Adjusted R Square	0.52842
Standard Error	41.33032
Observations	10

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	18934.9348	18934.9348	11.0848	0.01039
Residual	8	13665.5652	1708.1957		
Total	9	32600.5000			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	98.24833	58.03348	1.69296	0.12892	-35.57720	232.07386
Square Feet	0.10977	0.03297	3.32938	0.01039	0.03374	0.18580

According to these results, the regression equation is

*Predicted House Price = 98.24833 + 0.10977 * Area in square feet*

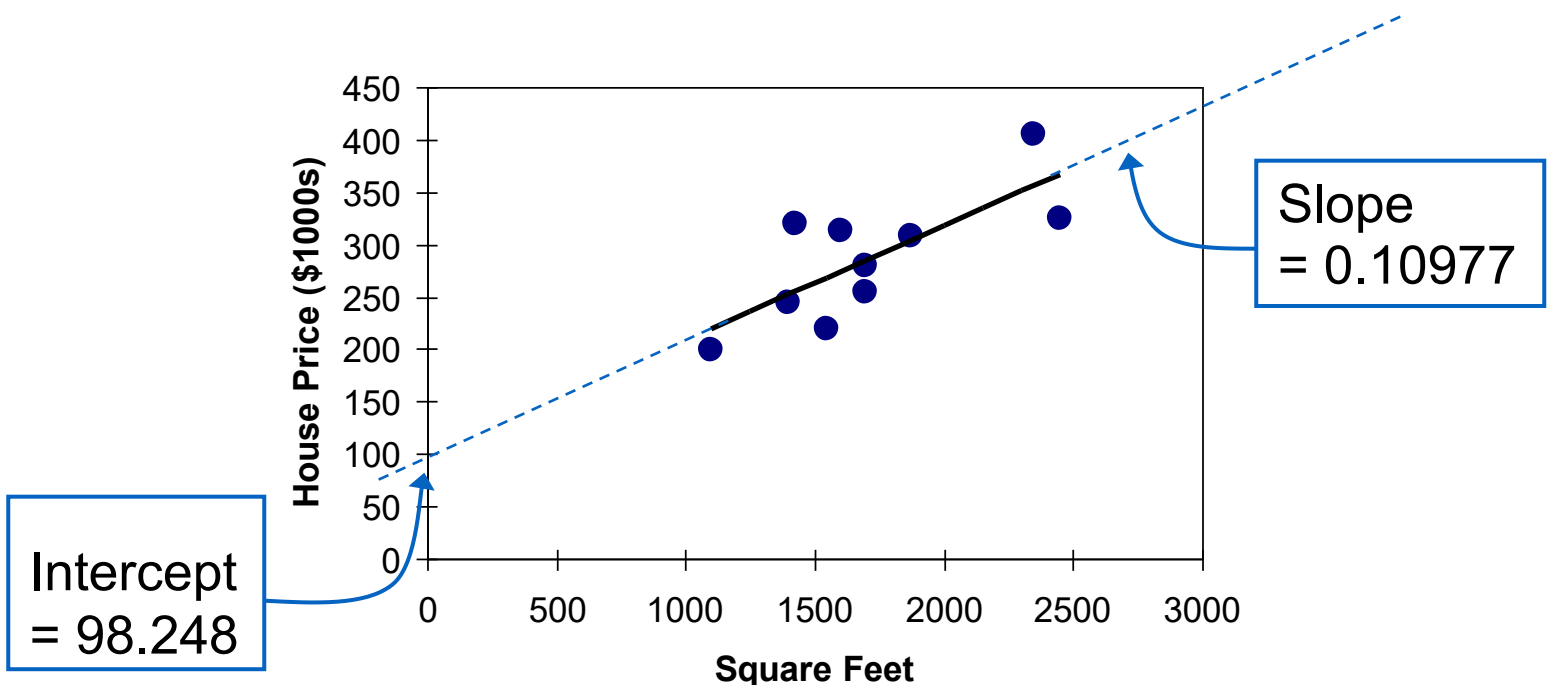
The R² is 0.58082.

The sums of squares, associated degrees of freedom, F-statistic, and its p-value are given in ANOVA. Here, $F=11.0848$, and the p-value is 0.01039 (always one-sided, right-tailed in the ANOVA F-test).

Coefficients (intercept $b_0=98.24833$ and slope $b_1=0.10977$) are followed by their standard errors, t-statistics for testing

$H_0: \beta_0=0$ and $H_0: \beta_1=0$, the corresponding p-values, and finally, 95% confidence intervals for β_0 and β_1 .

The toolpak can also produce the fitted line plot, if you check the corresponding box.



$$\text{house price} = 98.24833 + 0.10977 (\text{square feet})$$

Without the Data Analysis Toolpak, Excel has functions

= **SLOPE (A:A, B:B)** for the slope and = **INTERCEPT (A:A, B:B)** for the intercept.

Correlation

Excel function =**CORREL (column X, column Y)**

computes the correlation coefficient between X and Y.

The data analysis toolpak also has a function **Correlation**.

ex02-26beer.xls [Comp...

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW PAMFAX

From Access From Web From Text From Other Sources Existing Connections Refresh All Connections Sort Filter Clear Reapply Advanced Text Columns

A1 : PercentAlcohol

	A	B	C	D	E	F	G	H	I
1	PercentAlcohol	Calories							
2	4.10	136							
3	4.10	132							
4	3.20	96							
5	4.90	153							
6	4.20	95							
7	5.90	157							
8	4.10	94							
9	4.60	155							
10	5.20	177							
11	4.70	163							
12	4.70	154							
13	4.70	163							
14	4.80	153							
15	5.40	171							

Data Analysis

Analysis Tools

- Anova: Two-Factor Without Replication
- Correlation**
- Covariance
- Descriptive Statistics
- Exponential Smoothing
- F-Test Two-Sample for Variances
- Fourier Analysis
- Histogram
- Moving Average
- Random Number Generation

OK Cancel Help

The obtained correlation matrix contains correlation coefficients for each pair of variables.

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW

From Access From Web From Text From Other Sources Existing Connections Refresh All Connections Sort Filter

D1 :

	A	B	C	D	E	F
1	PercentAlcohol	Calories			PercentAlcohol	Calories
2	4.10	136		PercentAlcohol	1	
3	4.10	132		Calories	0.910489795	1
4	3.20	96				
5	4.90	153				